Reflections on Brittany MacIntyre's Seminar: Insights from the BIO-OMIX Omega-3 Biomonitoring Pilot Study

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The overarching goal of Brittany MacIntyre's pilot study on Biomonitoring of the Omega-3 Index was to explore whether there are urine metabolite biomarkers that can measure the omega-3 index (O3I). In her talk, Brittany discussed O3I, which measures the percentage of two long-chain omega-3 fatty acids, EPA and DHA in red blood cell membranes. Higher levels of O3I are associated with lower incidence of cardiovascular disease, congenital heart disease mortality, neuropsychiatric disorders, and all-cause mortality. Although our body converts alpha-linoleic acid (AHA) into EPA and DHA, dietary omega-3 is the most effective way to increase EPA and DHA levels in the body. Therefore, investigating less invasive methods for assessing O3I could help better inform individuals about their dietary needs in a more accessible way.

In her presentation, Brittany discussed the rationale behind the study's design and timeline. First, a preliminary pilot study was done to examine whether first-void urine could serve as an appropriate method for testing O3I levels rather than 24-hour urine. Although the gold-standard for testing substances in the body is a 24-hour urine collection, this requires clinical trial participants to carry around large jugs throughout their day for the length of the trial to collect their urine. As this is a cumbersome task that would drive potential participants away from contributing to their research, Brittany and her team did a secondary analysis of an existing clinical trial where they explored the use of first-void urine samples to test for urine metabolite biomarkers as it is much more accessible. From an initial analysis, her team found that S-Carboxypropylcysteamine (CPCA) found in first-void and 24-urine collection samples were highly correlated, meaning that further testing could be done using primarily first-void samples. After confirming that first-void urine could be used in an analysis of urine metabolite markers for detecting changes in O3I, Brittany and colleagues designed a study that would address (1) whether urinary biomarkers serve as acute or chronic biomarkers for omega-3 status, and (2) determine the specificity of urinary biomarkers to the O3I versus plasma EPA and DHA levels. From a time-response trial where participants were supplemented with omega-3, the team gathered data about acute and chronic changes of lipid-fractions in plasma and red-blood cells and matched the urinary metabolite changes to one of these components.

In this study, Brittany used several methods to conduct her analysis. The first was Partial Least Squares Discriminant Analysis (PLS-DA), which acts as a supervised principal components analysis (PCA) to cluster metabolites based on the time periods that characterized the study.

Interestingly, they were not able to find significant changes in CPCA, but did find that two other markers were significantly associated, an unknown dianion and tetrahydroaldosterone glucaronide (THA-g). Next, Lease Absolute Shrinkage and Selection Operator (LASSO) was used for feature selection to more directly determine which markers and factors are associated with changes in O3I and EPA and DHA. Brittany then used the unknown dianion and THA-g to build classifiers for high and low levels of omega-3 in plasma and red blood cells. A Linear Mixed Effects Model (LMM) was then fitted with participant variation as random-effects to determine how much variation was explained by the model.

During this seminar presentation, I learned that building a robust study design and remaining flexible in the analytical approach are fundamental aspects of bioinformatics research. I found it particularly interesting how Brittany described the detailed plans developed for assessing various components of omega-3 detection, and how unexpected challenges emerged once the study was underway. For example, when the team did not observe significant changes in CPCA, Brittany acknowledged the disappointment but emphasized the importance of continuing the analysis using the markers that did show meaningful associations with O3I and EPA/DHA percentages. Moving forward in my own work, I will adopt a similar mindset: designing thoughtful, well-structured workflows that do not depend on achieving a specific or expected result in order to proceed. While research cannot always avoid encountering unexpected outcomes, establishing systems that allow any result to be interpreted in an insightful and productive way is essential.

In addition to the overall study design, I learned that robust analyses are often built by validating each step and layering insights gradually. Brittany demonstrated this well by using several analytical tools and machine-learning algorithms, where each method incorporated findings from the previous one to deepen the overall examination. Although I was not familiar with some of the techniques she described, her thorough approach motivated me to learn more about the statistical, computational, and machine-learning tools she applied, such as LASSO, and how they might be useful in my own research interests, including characterizing biodiversity to support ecosystem preservation. Even though her talk was outside my primary area of focus, it helped me recognize how classifier tools and predictive modeling could be integrated into the types of analyses I hope to conduct.